

Impact of soil texture on temporal and spatial development of osmotic-potential gradients between bulk soil and rhizosphere

Doris Vetterlein^{1,2*}, Krisztian Szegedi^{1,2}, Florian Stange^{1,2}, and Reinhold Jahn¹

¹ Institute of Soil Science and Plant Nutrition, Martin-Luther-University Halle-Wittenberg, Weidenplan 14, 06108 Halle (Saale), Germany

² Department of Soil Ecology, UFZ – Helmholtz Centre for Environmental Research, Theodor-Lieser-Str. 4, 06120 Halle (Saale), Germany

Accepted September 20, 2006

Summary

Solute transport from the bulk soil to the root surface is, apart from changes in soil moisture and plant nutrient uptake, a prerequisite for changes in soil osmotic potential (Ψ_o). According to the convection-diffusion equation, solute transport depends on a number of parameters (soil moisture-release curve, hydraulic conductivity, tortuosity factor) which are functions of soil texture. It was thus hypothesized that soil texture should have an effect on the formation of Ψ_o gradients between bulk soil and the root surface. The knowledge about such gradients is important to evaluate water availability in the soil-plant-atmosphere continuum (SPAC).

A linear compartment system with maize grown under controlled conditions in two texture treatments (T1, pure sand; T2, 80% sand, 20% silt) under low and high initial application of salts (S1, S2) was used to measure the development of Ψ_o gradients between bulk soil and the root surface by micro-scale soil-solution sampling and TDR sensors.

The differences in soil texture had a strong impact on the formation of Ψ_o gradients between bulk soil and the root surface

at high and low initial salt application rate. At high initial salt application, a maximum osmotic-potential gradient ($\Delta\Psi_o$) of -340 kPa was observed for the texture treatment T2 compared to $\Delta\Psi_o$ of -180 in T1. The steeper gradients in osmotic potential in treatment T2 compared to T1 corresponded to higher cumulative water consumption in this treatment which can partly be explained by higher soil hydraulic conductivity in the range of soil matric potentials covered during the duration of the experiments. Differences between texture treatments in Ψ_o at the root surface did not result in differences in plant-water relations measured as gas-exchange parameters (transpiration rate, water-use efficiency) and leaf osmotic potential. If soil osmotic and matric potential are regarded as additive in calculating the driving force for water movement from the soil into the root, the observed differences in water flux between treatments cannot be explained.

Key words: root-soil interface / root water uptake / solute transport

1 Introduction

The presence of solutes in soil-pore water creates an osmotic potential (Ψ_o). (In this paper, the osmotic potential or its negative, the osmotic pressure, will be regarded as a property of a solution, although strictly spoken an isolated solution has no osmotic pressure. Yet the same solution when placed in an osmometer with a semipermeable membrane can manifest an osmotic pressure. For a detailed discussion and definition of osmotic potential, cf., Nobel (1999)). However, for osmotic water transport to occur in soils, a semipermeable membrane is required where ions are excluded yet water can cross freely. Only two mechanisms are identified where an Ψ_o gradient because of varying solute concentrations in soil can cause water to move in either liquid or vapor phase (Kelly and Selker, 2001).

Liquid-water transport can occur when anions are excluded from negatively charged soil particles, which act as a semipermeable membrane. This can be observed in clays due to their charged surfaces.

Water-vapor transport can occur in unsaturated soils when differing solute concentrations exist across gas-filled pores

separating soil solution. The gas-liquid interfaces act as semipermeable membranes across which water vapor is transported.

These above mentioned mechanisms are only relevant in clay-rich soils or under conditions of high solute concentrations and low water content. For most cases of practical interest, the effect of salt concentration gradients on the driving force for water movement in soil is small enough to be disregarded (Hillel, 1998, p. 256).

However, the osmotic effect is important in the interaction between plant roots and soil (Hillel, 1998, p. 151). In the classical concept, roots are regarded as nearly perfect osmometers, the endodermis with its casparian bands is supposed to act as the “root membrane”. When solute is completely restricted by the semipermeable membrane, water flow in response to a difference in Ψ_o is equivalent to that caused by a hydraulic potential difference of equal magnitude, and the two driving forces are therefore additive and act through the same transmission coefficient K (hydraulic conductivity) (Hillel, 1998).

To investigate the relative importance of Ψ_o in relation to soil matric potential (Ψ_m), a method was developed to measure the temporal and spatial dynamics of Ψ_o in the vicinity of roots (Vetterlein and Jahn, 2004a). This method allowed cor-

* Correspondence: Dr. D. Vetterlein; e-mail: doris.vetterlein@ufz.de

roborating the hypothesis of *Stirzaker and Passioura* (1996) that salts may accumulate at the root surface and, thus, Ψ_o at the root surface may be more negative compared to the bulk soil. It has been shown that the initial salt concentration in the substrate (*Vetterlein and Jahn*, 2004b) as well as the type of salt in combination with specific uptake mechanisms of different genotypes (*Vetterlein et al.*, 2004) can affect the formation of such gradients in Ψ_o between bulk soil and the root surface.

However, apart from the initial salt concentration and plant nutrient uptake, nutrient transport in the soil is an important impact factor for the ion concentration at the root surface and the resulting osmotic potential in the soil solution at the root surface.

Solute transport from the bulk soil to the root surface is described by the convective-diffusion equation (*Darrah and Roose*, 2001). Equation 1 is presented in a modified form, describing the planar geometry of the compartment system approach:

$$\frac{\partial((b + \theta)C_i)}{\partial t} = \frac{\partial}{\partial x} \left(D_w \theta f \frac{\partial C_i}{\partial x} \right) - \frac{\partial}{\partial x} (qC_i), \quad (1)$$

where C_i is the concentration in soil, t is the time, D_w is the diffusion coefficient in water, x is the distance from the bulk soil to the root surface, θ is the soil water content, q is the water flux, b is the buffer power, and f is the tortuosity factor.

Both terms of Eq. 1, the first one representing diffusion (Eq. 2), the second one representing convection (Eq. 3), contain parameters which depend on soil texture:

$$\frac{\partial}{\partial x} \left(D_w \theta f \frac{\partial C_i}{\partial x} \right), \quad (2)$$

$$q = -K(\Psi_m) \frac{\partial \Psi_m}{\partial x}. \quad (3)$$

K is the soil hydraulic conductivity, and Ψ_m is the soil matric potential.

The substrate-specific soil moisture–release curve $\theta(\Psi_m)$, the hydraulic conductivity K , and the tortuosity factor f are all functions of pore-size distribution and continuity of pores in soil.

It is thus hypothesized that soil texture in addition to initial salt content and plant uptake characteristics should have an effect on the formation of Ψ_o gradients between bulk soil and the root surface.

So and Nye (1989) specifically investigated the importance of soil water content and bulk density for the diffusion coefficient and tortuosity factor f for two different soils varying in texture. They concluded that water content and matric potential were the main factors controlling f with bulk density making a small contribution. Soil type had also a significant effect on f . In pot experiments, *Cox and Barber* (1992) and *Seiffert et al.* (1995) have shown for P and K that diffusion in soils is strongly dependent on soil water content, soil type, and bulk density.

However, differences in soil moisture may also have an impact on root growth and thus have an indirect effect on nutrient uptake (*Barber and Mackay*, 1985). Care has to be taken in selecting the experimental design to allow differentiation between the two mechanisms (diffusion coefficients, root growth). Compartment systems can be used for such a purpose as they are based on the formation of a root mat, which is regarded as the root surface, disregarding differences in root-length density, root distribution, and root morphology (*Dunham and Nye*, 1973).

From Eq. 3, it is obvious, that not only diffusion but also convection is dependent on pore-size distribution as soil hydraulic conductivity $K(\theta)$ has a major impact on water flux (convection) from the bulk soil to the root surface. This is true irrespective of the controversial debate whether the major resistance to water transport in soil-plant-atmosphere continuum (SPAC) is located in the soil or in the plant (*cf.*, summaries in *Bristow et al.*, 1984; *Hillel*, 1998; *Kramer and Boyer*, 1995).

Hacke et al. (2000) have shown the impact of different soil textures on plant water use for loblolly pine plantations on loam versus sand. Again, parts of the limitations posed by soil texture were overcome by shifts in root distribution. Similar observations were reported by *Li et al.* (2005) for cotton, investigating the ratio between root length and leaf area on substrates varying in texture.

The complex interaction of soil texture, plant hydraulic conductance, rooting depth, and soil matric potential in respect to water flow, is described in detail by a modeling approach in *Jackson et al.* (2000).

Varying soil texture and thus soil moisture–release curve and $K(\theta)$ might provide a tool for investigating the impact of decreasing Ψ_o at the root surface on plant water uptake without altering the nutrient status of the plants. This is a problem which cannot be ruled out in experiments varying the initial amount of salt in the substrate or varying the type of salt in the substrate as they are often conducted in investigating the effect of osmotic potential on plant growth or water uptake (*Sonneveld et al.*, 2004).

In order to change soil physical properties without changing soil chemical properties, substrates were mixed from quartz of different particle-size classes. As quartz has only very little sorption sites, nutrient concentrations in such substrates are highly favoring the importance of convection over diffusion (*Claassen*, 1990). The experiment was conducted two times, one time at low initial salt content (S1) and a second time at high initial salt content (S2). Studies were conducted in a linear compartment model system using *Zea mays*.

The aim of the present investigation was to determine in a model system the impact of soil texture on the dynamics and extend of Ψ_o gradients between bulk soil and the root surface and to verify whether differences in Ψ_o at the root surface will result in differences in plant-water relations between the treatments.

2 Materials and methods

2.1 Quartz substrates with different soil physical characteristics

Substrate T1 consisted of a quartz sand (C horizon of a Regosol, Köselitz, Germany) sieved to <1 mm. Substrate T2 was derived from mixing T1 (80%) with a quartz silt (20%) (Mikrosil SP12, Westdeutsche Quarzwerke GmbH).

The soil moisture–release curves, based on a suction plate/pressure plate method, were established for both substrates using the program RETC (US Salinity Laboratory, Riverside CA). This is based on the equation of *van Genuchten* (1980) and the restriction $m = 1 - 1/n$:

$$\theta = \theta_r + \frac{\theta_s - \theta_r}{[1 + (\alpha|\Psi_m|)^n]^m} \quad (4)$$

The pure quartz sand (T1) had a bulk density of 1.3 g cm^{-3} , substrate T2 (80% quartz sand plus 20% quartz silt) had a bulk density of 1.45 g cm^{-3} . (The two treatments T1 and T2 are referred to as different-texture treatments although they differed also in bulk density. However, the differences in bulk density resulted from the differences in texture.) Substrates were handled in the same manner, *i.e.*, during packing of the compartment model system, soil moisture content was 1% (w/w), and the same force was applied in joggling the boxes. As a result of differences in bulk density, the total soil volume differed between the treatments by 10.5%.

The saturated hydraulic conductivity (K_s) was measured with constant-head permeameter (*Klute and Dirksen*, 1986) in six replicates for each substrate and compared to estimates derived from the Program Rosetta Lite V. 1.0 (*Schaap et al.*, 2001). Measurements were problematic as bulk density in T2 increased during measurement, which was not the case in experiment S1 and S2 (detailed description see below) for either of the treatments. Data on hydraulic conductivity K are based on soil moisture–release curves established with RETC using the approach of *Mualem* (1976):

$$\frac{K(\Psi_m)}{K_s} = \frac{\left[1 - \left(1 - \frac{1}{1 + (\alpha|\Psi_m|)^n}\right)^{-m}\right]^2}{[1 + (\alpha|\Psi_m|)^n]^{m/2}} \quad (5)$$

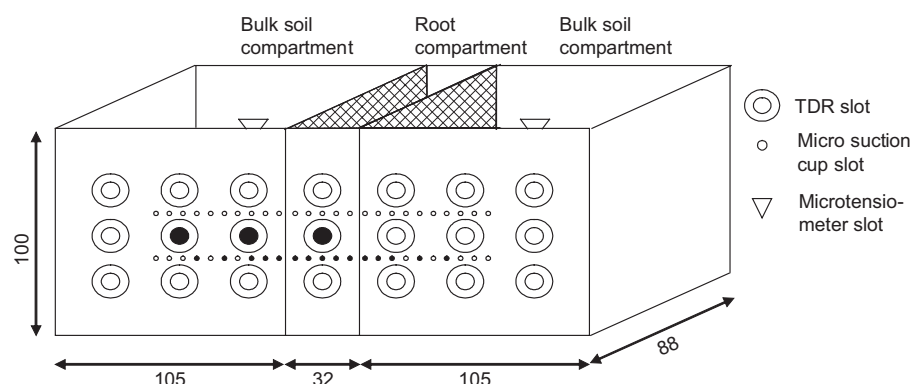


Figure 1: Sketch of compartment system. Dimensions are given in mm. Slots actually equipped with sensors in the experiment are marked in black. Microtensiometers are mounted from the back, exactly opposite the TDR sensors. A hydrophilic nylon net (hatched) of 30 μm mesh separates the root compartment from the bulk soil compartment. Modified from *Vetterlein and Jahn* (2004a).

2.2 Experimental design

Zea mays L. cv. Helix was grown in boxes (one plant per box) under controlled conditions (20°C , 60% RH, and a 12 h photoperiod) (*Vetterlein and Jahn*, 2004a). The root compartment was separated from the bulk soil compartments by nylon nets (30 μm mesh). The root mat formed along the nylon net is regarded as the root surface. Microtensiometers, TDR probes, and micro-suction cups were aligned horizontally, at increasing distance from the center of the root compartment, with a spatial resolution of 30 mm, 30 mm, and 6 mm, respectively (*Vetterlein and Jahn*, 2004a; Fig. 1). Each compartment system was equipped with three microtensiometers, three TDR probes, and 15 micro-suction cups. The experiment was set up with three replications as a randomized block design.

Two separate experiments were conducted, one at a low initial salt content (S1) and one at a high initial salt content (S2).

In experiment S1, both substrates (T1, T2) were initially supplied with 100 mg N kg^{-1} (NH_4NO_3), 80 mg P kg^{-1} (CaHPO_4), $100 \text{ mg Ca kg}^{-1}$ (CaSO_4), 100 mg K kg^{-1} (K_2SO_4), 50 mg Mg kg^{-1} (MgSO_4), and a micronutrient solution ($3.25 \text{ mg Mn kg}^{-1}$, $0.79 \text{ mg Zn kg}^{-1}$, $0.5 \text{ mg Cu kg}^{-1}$, and $0.17 \text{ mg B kg}^{-1}$).

In experiment S2, fertilizer application was similar to S1, but the amount of K and Mg fertilization was doubled.

Compartment systems were moistened by capillary rise to similar initial soil matric potential of -2.5 kPa with distilled water, by placing the compartment system on a column of quartz sand (particle size 1–2 mm), 20 cm in height, which was filled with water. After the initial watering was finished, water was removed from the quartz column by suction. Thereafter, system equilibrated for 6 d until germination was observed. Micro-suction cups were installed 1 d before germination was observed (= 0 DAP).

Soil solution was sampled 7, 24, 28, 31, 35, 38, 42, 45, 49, 52, and 55 days after planting (DAP) in experiment S1 and 7, 18, 21, 25, 28, 32, 35, and 39 DAP in experiment S2. Solution from the first sampling date was discarded. Up to 38 and 28 d after planting in S1 and S2, respectively, the soil solution was collected under a suction of 30 kPa for 15 min on the respective day. Thereafter, suction was increased according to

decreasing soil Ψ_m , and the sampling interval was extended to obtain at least 15 μL of solution. Microtensiometer and TDR-probe measurements were taken every 30 min throughout the duration of the experiments.

During the course of the experiment S2, leaf area was estimated by measuring length and width of each leaf at the dates of soil-solution sampling. These estimates were calibrated against true leaf area measured by destructive sampling at harvest.

Plant water uptake was calculated from changes in soil matric potential in root and bulk soil compartment. Calculation of cumulative water uptake was based on day 11 after planting. Days 0 to 11 were not taken into account. During this period, plants were still very small, and changes in soil matric potential were in the range of 0.5 kPa while the precision of microtensiometer measurement is only 0.25 kPa. However, the water-retention curve is very steep for soil matric potentials between -2.5 and -3.5 kPa (Fig. 2) and, thus, the uncertainty of water-uptake calculations for this period is very high.

Dry weight of plant samples was determined at harvest (55 DAP in experiment S1, 40 DAP in experiment S2) after drying at 65°C for 48 h. Thereafter, ground plant samples were digested with HNO_3 in a pressure unit (Seiff), and P, K, Mg, and Ca were determined by ICP-OES (Jobin Yvon). Nitrogen content was analyzed by furnace combustion with a CHN analyzer (Varian). The soil was sampled in slices of increasing width with increasing distance from the nylon mesh (4, 8, 16, 32 mm). In addition, one soil sample was taken from the root compartment, and one sample was collected by brushing off the small amount of soil adhering to the nylon nets. The oven-dry soil (105°C) was extracted with distilled water (1:25) for 2 h. Concentrations of K^+ , Mg^{2+} , Ca^{2+} , and SO_4^{2-} were measured in the supernatant after centrifugation and filtration.

The soil solution Ψ_o was measured cryoscopically requiring a volume of 15 μL (Osmomat 300, Gonotec GmbH).

For the measurement of soil osmotic potential with TDR-sensors, calibration was conducted as described in Vetterlein and Jahn (2004a).

Plant-water relations were determined by measuring leaf gas exchange with HCM-1000 (Walz GmbH) using an additional light source ($270 \mu\text{mol m}^{-2} \text{s}^{-1}$) and measuring leaf Ψ_o in the press sap of leaves. Gas-exchange measurements were conducted *in situ* 28, 29, 32, 34, 35, 36, and 39 d after planting in experiment S2. For two of the three replications, gas exchange was measured for 2 h on the respective day. For leaf Ψ_o measurement, a leaf part of comparable physiological age was cut off 24, 31, 38, 45, 52, and 55 d after planting in experiment S1 and 21, 28, 32, 35, and 39 d after planting in S2.

Differences between treatments were tested by one-factorial ANOVA (SPSS, Version 9.0) at $p < 0.05$. Standard deviations are given in all graphs except for gas-exchange parameters.

2.3 Planar versus cylindrical system

The compartment system used in the present experiments represents a planar rhizobox system which is a standard tool in rhizobox research (Hinsinger et al., 2005; Wenzel et al., 2001) despite the altered geometry compared to a single cylindrical root. Although altered geometry has an influence on the development of concentration gradients, it has been shown by Kirk (1999) that the results of radial and planar geometries converge as the root radius chosen for model calculations increases. The planar system was chosen, not only to avoid that effects of texture and root growth are confounded (see above), but especially as it enables measurements with high spatial resolutions with sensors which could not be used to measure along a single root. Temporal resolution of soil-solution sampling is too low to measure gradients along single roots growing with a velocity of $2\text{--}3 \text{ cm d}^{-1}$ as soil-solution equilibrium is disturbed by sampling (Vetterlein and Jahn, 2004a). TDR-sensors are 5 cm long, and the required spatial

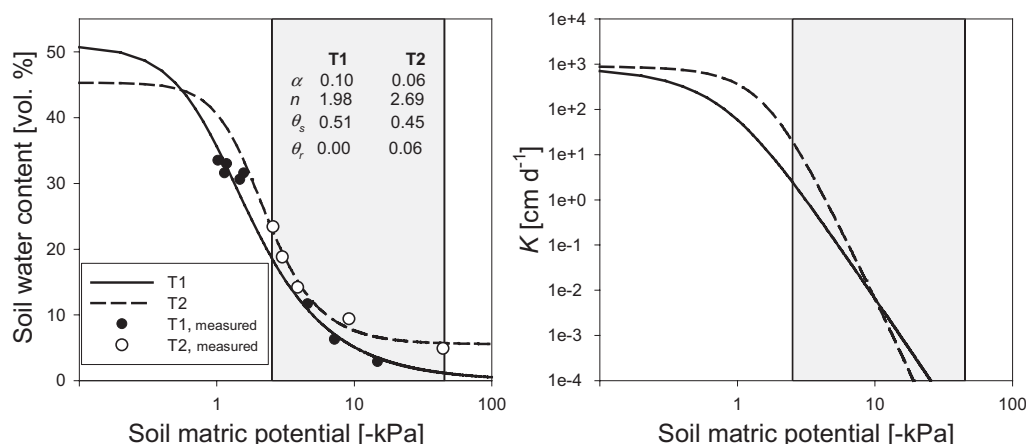


Figure 2: Soil moisture–release curve and hydraulic conductivity for substrate T1 (100% quartz sand, bulk density of 1.3 g cm^{-3}) and substrate T2 (80% quartz sand, 20% quartz silt, bulk density of 1.45 g cm^{-3}). The fitted parameters α , n , θ_s , and θ_r for the calculation of $\theta(\Psi_m)$ and $K(\Psi_m)$ are given for both substrates. The grey area represents the range of matric potentials covered in experiments S1 and S2. Measured points for matric potential < -20 kPa are taken from a separate calibration experiment (Vetterlein and Jahn, 2004a).

resolution can only be obtained once they can be aligned parallel to the object of interest as is the case for the root mat.

In the present work, the term gradient refers to changes of a parameter with increasing distance from the nylon net, which is regarded as the root surface. The root compartment itself is assumed to be homogeneous, which is confirmed by the data measured with five suction cups across the root compartment shown in Fig. 3 for the osmotic potential.

3 Results and discussion

Differences in texture resulted in higher pore volume and lower residual water content in T1 compared to T2 (Fig. 2). However, in the range of soil matric potentials covered during the duration of the experiments (–2.5 to –45 kPa), soil water content was always higher in T2 compared to T1 for the same matric potential. This is not the case for the unsaturated hydraulic conductivity, which is higher in T2 compared to T1 in the range of –2.5 to –10 kPa but equal or lower in T2 compared to T1 at lower soil matric potentials.

The point of intersection between the $K(\Psi_m)$ curves for the two treatments depends on the estimates used for K_s and on the parameters α and n . We have chosen a K_s value of 890 cm d^{–1} for both substrates in Fig. 2. This is the value measured for substrate T1. If a value <890 cm d^{–1} is selected for K_s of T2, the point of intersection moves to the right and *vice versa* if a larger value is chosen. It has to be noticed,

that the parameter α and n in the van Genuchten equation (Eq. 4) are afflicted with great uncertainty if there are only a few data points for matric potential and soil water content in the range of 0 to –2 kPa as it is the case if the water-retention curve is determined by standard procedures (Klute and Dirksen, 1986). This is of special relevance in sandy substrates, which show a steep slope of the water-retention curve in this range. For the water-retention curve itself, this is not a problem as long as there is good agreement of calculated and measured values in the matric-potential range of interest. However, the uncertainty is carried over to the determination of K according to Eq. 5 as this is based on the same parameters α and n . Despite the uncertainty in the determination of K , the data show how differences in K can develop with changing water content and how these differences might affect transport (Fig. 4).

The differences in soil texture had a strong impact on the formation of Ψ_o gradients between bulk soil and the root surface irrespective of initial salt application (Fig. 3). However, gradients were steeper with higher initial salt application (experiment S2). Gradients started to develop ca. 15–20 d after planting (Fig. 3, Fig. 5) in experiment S2. At this time, cumulative water uptake started to deviate between treatments, and hydraulic conductivity was still slightly larger in T2 compared to T1.

Differences in extent of Ψ_o gradients between bulk soil and the root surface for the treatments T1 and T2 cannot be explained by differences in matric-potential gradients be-

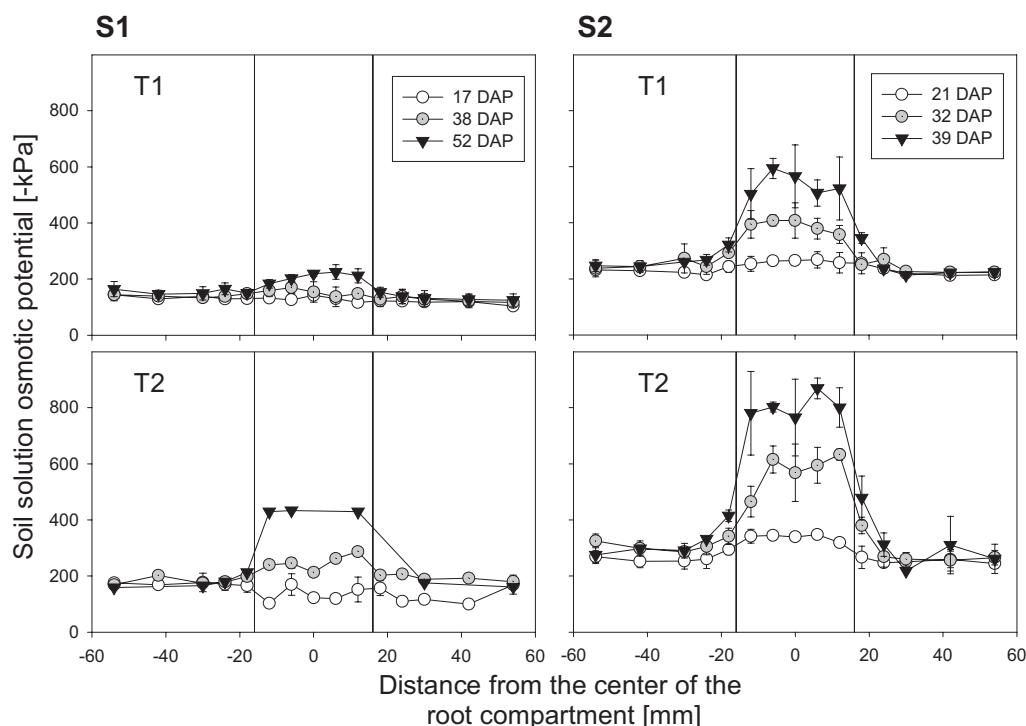


Figure 3: Change of Ψ_o with time (17, 38, 52 days after planting (DAP) in S1 and 21, 32, 39 DAP in S2) and with increasing distance from the center of the root compartment. Data are presented for two texture treatments (T1, T2) in two experiments (S1, S2) differing in initial salt application. Vertical lines indicate the position of the nylon net. The standard deviations between the three replicates of the treatments are indicated by vertical bars.

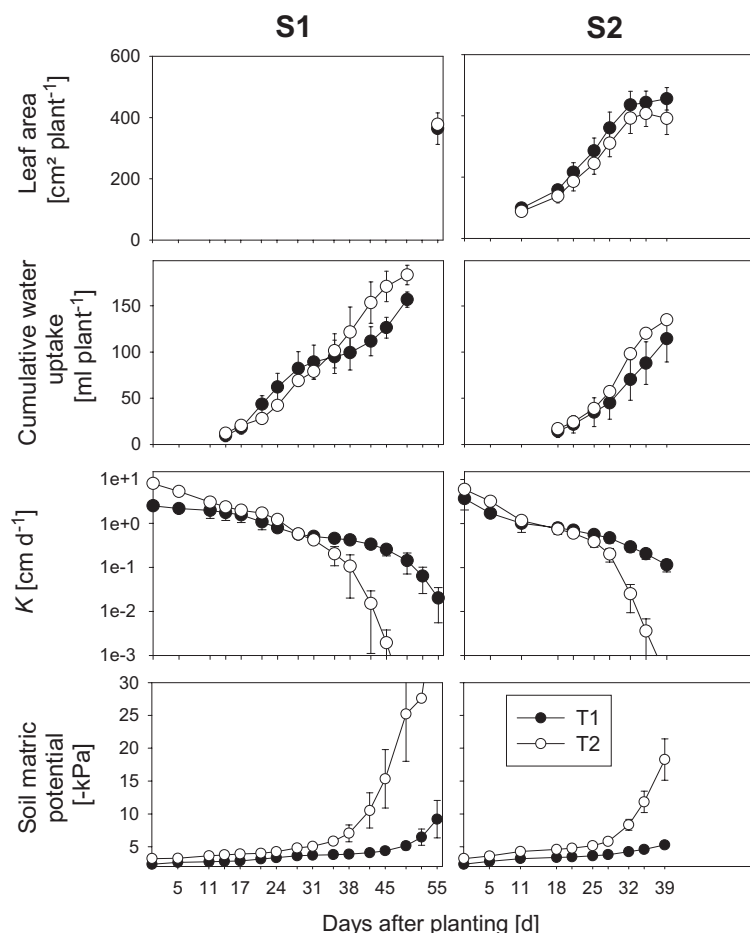


Figure 4: Change of leaf area, cumulative water uptake, soil hydraulic conductivity (K), and soil matric potential in the root compartment with time for two texture treatments (T1, T2) in two experiments (S1, S2) differing in initial salt application. Bars represent standard deviation of the three replications for each treatment. The calculation of cumulative water uptake is based on day 11 after planting. The 10.5% difference in volume between treatments T1 and T2 is taken into account.

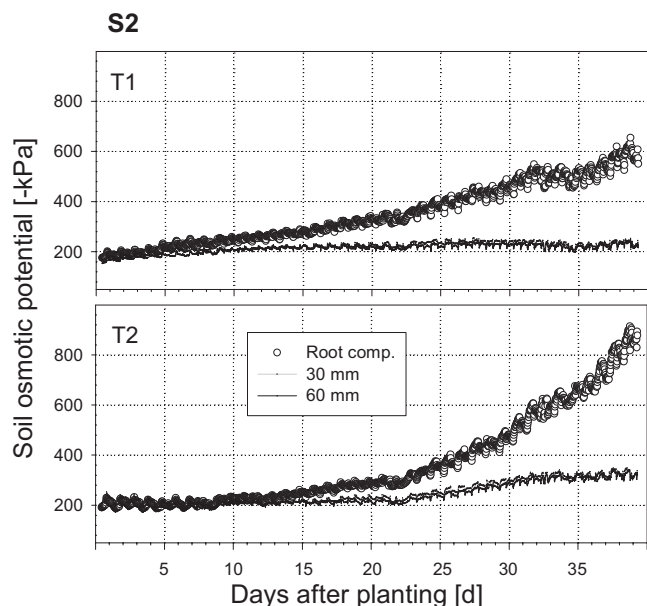


Figure 5: Change of Ψ_o with time for the root and bulk soil compartment for one of the three replicates of treatment T1, T2. Data are derived from TDR measurements using the regression deduced in Vetterlein and Jahn (2004a). As no differences occurred between Ψ_o at 30 and 60 mm distance from the root compartment, the values overlay each other.

tween bulk soil and root compartment. A gradient in soil matric potential between bulk soil and root compartment was only measured in S1 for the last sampling date.

Differences in the extent of Ψ_o gradients between bulk soil and the root surface were confirmed by destructive soil sampling at harvest (Fig. 6). In both experiments, significantly higher K and Mg concentrations were found in the root compartment and at the nylon nets in treatment T2 compared to T1.

Differences in accumulation in the root compartment can partly be explained by differences in plant nutrient uptake (Tab. 1). Roots were not analyzed due to the problem, that especially for non-cell wall constituents like potassium (K), it is difficult to clean roots sufficiently without losing part of the nutrient in question (Grzebisz et al., 1989).

Differences in nutrient uptake into the shoot were 11 and 13 mg for K in S1 and S2, respectively. Differences in K accumulation in the soil between treatments sum up to 49 and 63 mg for S1 and S2, respectively, for the root compartment and the bulk soil up to a distance of 60 mm from the root surface. The same is true for Mg for which the differences in accumulation in the soil are 13.8 and 25.4 mg for S1 and S2, respectively. The residual soil fraction 60–105 mm from the nylon net was not analyzed in the present experiment.

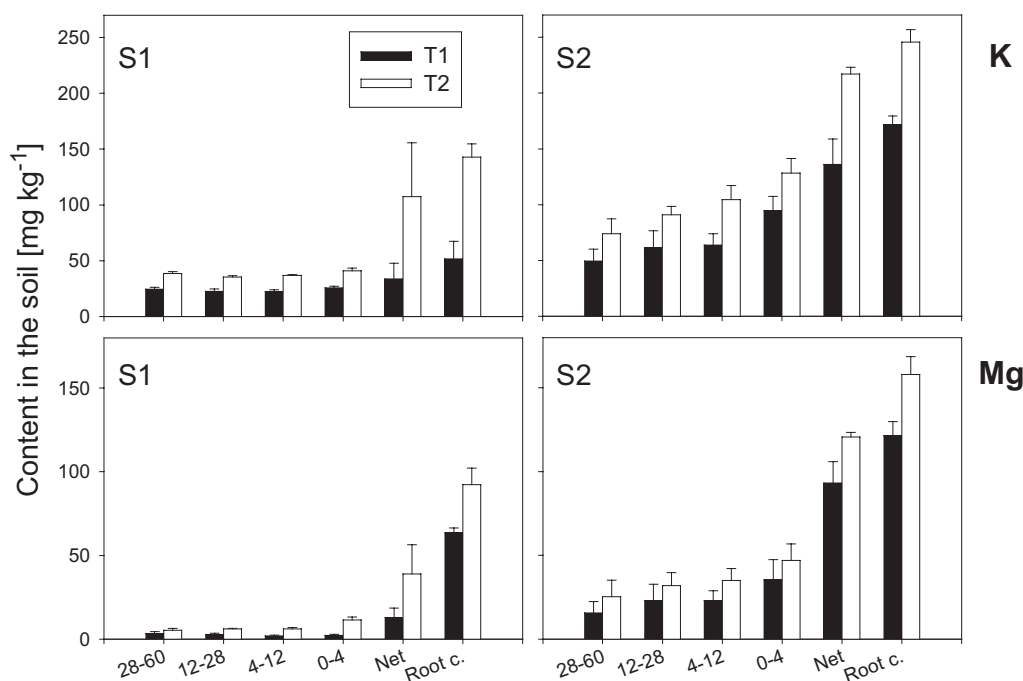


Figure 6: Impact of texture treatment (T1, T2) on the distribution of K and Mg in the soil at harvest (55 and 39 DAP) with increasing distance from the root compartment in two experiments (S1, S2) differing in initial salt application. Bars represent standard deviation of the three replications for each treatment.

Table 1: Impact of texture (T1, T2) on plant dry weight, leaf area, leaf nutrient content, and uptake into the shoot at harvest (39 and 55 days after planting, respectively) at low and high initial salt application (S1, S2). Standard deviation is given in brackets. * indicates significant difference between treatments ($p < 0.05$).

Parameter	S1		S2	
	T1	T2	T1	T2
Shoot dry weight [g plant ⁻¹]	3.3 (0.5)	2.4 (0.2)	1.8 (0.3)	1.5 (0.3)
Root dry weight [g plant ⁻¹]	1.5 (0.4)	1.1 (0.2)	0.6 (0.2)	0.6 (0.1)
Leaf area [cm ² plant ⁻¹]	676 (163)	526 (96)	456 (37)	391 (52)
Nutrient content in leaves [mg g ⁻¹]				
N	27.8 (1.1)	31.5 (0.2)	34.2 (1.4)	36.0 (1.3)
P	7.2 (1.4)	6.5 (0.4)	7.3 (0.9)	6.7 (0.5)
K	26.3 (2.5)	31.2 (0.4)	35.3 (2.1)	36.3 (1.2)
Ca	3.7 (0.2)	2.4 (0.2)	1.2 (0.2)	1.0 (0.2)
Mg	3.4 (0.2)	2.4 (0.1)	2.6 (0.2)	2.3 (0.1)
Nutrient uptake into the shoot [mg plant ⁻¹]				
N	101 (11.2)	87.9 (5.4)	72.2 (10.8)	61.9 (12.6)
P	26.6 (1.9)	17.6 (1.9)	14.3 (0.8)	11.2 (1.7)
K	103 (7.5)	91.9 (5.6)	86.6 (10.4)	74.3 (13.7)
Ca*	14.1 (2.4)	6.8 (0.8)	2.4 (0.1)	1.8 (0.1)
Mg	14.0 (2.6)	7.5 (0.6)	5.9 (1.3)	4.2 (1.1)

Differences in nutrient accumulation or Ψ_o gradients between bulk soil and the root surface correspond to differences in cumulative water uptake which can be regarded as the water flux q and was higher in treatment T2 compared to treatment T1 in both experiments (Fig. 4).

Several factors might have contributed to the slightly higher water consumption in treatment T2 compared to T1. Leaf area (Fig. 4) did not differ significantly between treatments in neither of the two experiments. This is not only true for the

harvesting date but also for the time course of leaf area development.

There are numerous reports (Mengel and Arneke, 1982; Peoples and Koch, 1979; Robin et al., 1989) on the impact of K nutritional status on stomatal regulation and thus transpiration rate and water-use efficiency of photosynthesis. According to Hsiao and Läuchli (1986), stomatal conductivity increases with increasing K supply, however, in some plant species like maize, an optimal K level is reached where maxi-

mum stomatal conductivity is observed and stomatal conductivity can decrease at very high K levels which are not yet regarded as toxic.

There was a large difference in the K supply between experiment S1 and S2 which is reflected in leaf K concentrations at harvest (Tab. 1). In stalks, K concentrations up to 65 mg kg⁻¹ were measured, indicating luxury supply (data not shown). However, within each experiment, no significant differences in K nutritional status were observed between treatments T1 and T2. It is thus unlikely that K nutritional status contributed significantly to the differences in water consumption between the texture treatments in either of the experiments.

Leaf-gas exchange parameters, which were only measured in experiment S2, showed slightly higher transpiration rates in treatment T2 corresponding to a lower photosynthetic water-use efficiency (Fig. 7).

Leaf Ψ_o , which is a measure of leaf water status and forms together with leaf turgor the leaf water potential, was always more negative than soil-solution Ψ_o and decreased with time as soil matric potential decreased, indicating increasing

drought stress as has been shown in numerous cases (Hare et al., 1998; Morgan, 1984; Martinez et al., 2004).

However, leaf Ψ_o did not differ significantly between treatments T1 and T2 in neither of the two experiments whether measured at actual leaf water content (Fig. 8) or referred to fully turgid state (data not shown). This is in contrast to results from hydroponic studies where decreases in leaf osmotic potential with decreasing osmotic potential of the nutrient solution have been observed (Ackerson and Youngner, 1975; Bernstein, 1961). The lack of differences in leaf osmotic potential between treatments T1 and T2 despite the measured differences in rhizosphere osmotic potential cannot be attributed to marked differences in the “rate of development of water deficit” which is an important factor for osmotic adjustment as pointed out by Basnayake et al. (1996).

Looking at Eqs. 1 and 3, it is plausible that higher K values for the same matric potential in T2 compared to T1 resulted in higher water flux from the bulk soil to the root surface. However, as the determination of $K(\Psi_m)$ is afflicted with great uncertainty as discussed above, there is no proof for this hypothesis.

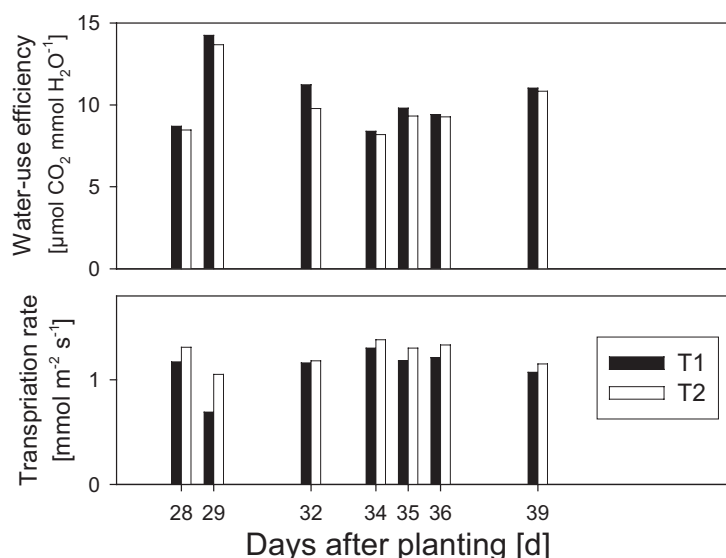


Figure 7: Change of transpiration rate and photosynthetic water-use efficiency with time for two texture treatments (T1, T2) in experiments S2 (high initial salt application).

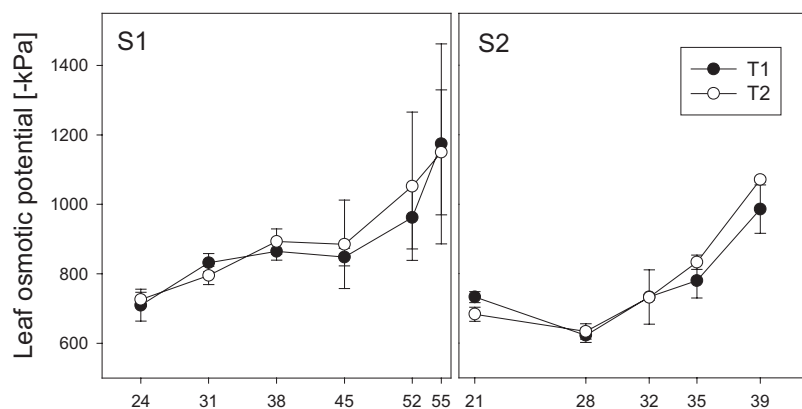


Figure 8: Change of leaf osmotic potential with time for two texture treatments (T1, T2) in two experiments (S1, S2) differing in initial salt application. Bars represent standard deviation of the three replications for each treatment.

At the start of the experiment, matric potential was similar for the two texture treatments as water was supplied to the compartments by capillary rise. However, during the time course of the experiment, matric-potential decline was stronger in T2 compared to T1. Measurable gradients in soil matric potential between root and bulk soil compartment, *i.e.*, $|\Delta\psi_m| > 0.2$ kPa, could not be detected with the exception of DAP 53–55 in treatment T1 in experiment S1. Thus, it can be assumed that in both texture treatments, hydraulic conductivity was high enough in relation to plant water requirement to avoid the development of gradients in soil matric potential between the bulk soil and the root compartment throughout the duration of both experiments.

As plants were grown under the same environmental conditions and leaf osmotic potentials indicated similar plant water potentials, loss of turgor was only observed at harvest, it is possible that differences in soil hydraulic conductivity were contributing to the differences in water flux in SPAC between the treatments.

Water flux in SPAC can be described as $q_{SPAC} = -\Delta\psi_{SPAC} \times K_{SPAC}$. Soil hydraulic conductivity together with the hydraulic conductivity of the plant forms K_{SPAC} .

$\Delta\psi_{SPAC}$, which is the water potential gradient between the plant and the soil, should be slightly smaller for T2 compared to T1 as soil matric potential was more negative in T2 compared to T1. However, as the value of matric potential is small compared to the value of plant water potential (Fig. 4, 8), this difference should have very little impact on water flux. This aspect is changing if soil ψ_o (Fig. 3, 5) is taken into account in an additive manner as suggested by Wadleigh and Ayers (1945) for water transport from the soil into the root. For water transport in the soil, ψ_o is not relevant as there are no semipermeable membranes at which osmotic-potential gradients could become effective. Soil ψ_o in experiments S1 and S2 were one order of magnitude lower than soil matric potentials. Thus, the differences in soil ψ_o at the root surface between treatments T1 and T2 should have a strong impact on water flux in SPAC, resulting in a smaller flux in T2 compared to T1 which is not in agreement with the observed cumulative water uptake. Thus, the results of the present experiments add to the controversial debate whether soil-solution osmotic potential and matric potential are really additive in respect to plant water availability (Wadleigh and Ayers, 1945; Hao and De Jong, 1988; Deanknox et al., 1998; Homae, 1999).

Acknowledgments

This work was supported by the *Deutsche Forschungsgemeinschaft (DFG)*. We gratefully acknowledge *Hans-Jörg Vogel, Andrea Schnepf, Hartmut Tanneberg, Gabriele Polcher, and Bodo Hofmann* for valuable comments on the script and technical assistance. We also owe a debt of gratitude to the precision engineering workshop of the *Chemistry Department of Martin-Luther-University* for constructing the compartment systems.

Abbreviations

C_i	concentration of a certain element in soil
D_w	diffusion coefficient in water
D_e	effective diffusion coefficient
f	tortuosity factor
K	soil hydraulic conductivity
K_s	saturated soil hydraulic conductivity
q	water flux
b	buffer power
t	time
x	distance from the bulk soil to the root surface
θ	soil water content, r residual, s saturated
ψ_m	matric potential
ψ_o	osmotic potential
ψ_{soil}	soil water potential
DAP	days after planting
SPAC	soil-plant-atmosphere continuum
TDR	time-domain reflectometry
S1	low initial salt application (100 mg K (kg soil) ⁻¹ , 50 mg Mg (kg soil) ⁻¹)
S2	high initial salt application (200 mg K (kg soil) ⁻¹ , 100 mg Mg (kg soil) ⁻¹)
T1	100% quartz sand
T2	80% quartz sand, 20% quartz silt

References

- Ackerson, R. C., Youngner, V. B. (1975): Responses of bermuda-grass to salinity. *Agron. J.* 67, 678–681.
- Barber, S. A., Mackay, A. D. (1985): Sensitivity analysis of the parameters of a mechanistic mathematical model affected by changing soil moisture. *Agron. J.* 77, 528–531.
- Basnayake, J., Cooper, M., Henzell, R. G., Ludlow, M. M. (1996): Influence of rate of development of water deficit on the expression of maximum osmotic adjustment and desiccation tolerance in 3 grain-sorghum lines. *Field Crops Res.* 49, 65–76.
- Bernstein, L. (1961): Osmotic adjustment of plants to saline media. I. steady state. *Am. J. Bot.* 48, 909–918.
- Bristow, K. L., Campbell, G. S., Calissendorff, C. (1984): The effects of texture on the resistance to water movement within the rhizosphere. *Soil Sci. Soc. Am. J.* 48, 266–270.
- Claassen, N. (1990): Nährstoffaufnahme höherer Pflanzen aus dem Boden. Severin Verlag, Göttingen, p. 327.
- Cox, M. S., Barber, S. A. (1992): Soil phosphorus levels needed for equal P uptake from four soils with different water contents at the same water potential. *Plant Soil* 143, 93–98.
- Darrah, P. R., Roose, T. (2001): Modeling the Rhizosphere, in Pinton, R.: The Rhizosphere. Marcel Dekker, New York, pp. 327–372.
- Deanknox, D. E., Devitt, D. A., Verchick, L., Morris, R. L. (1998): Physiological response of two turfgrass species to varying ratios of soil matric and osmotic potentials. *Crop Sci.* 38, 175–181.
- Dunham, R. J., Nye, P. H. (1973): The influence of soil water content on the uptake of ions by roots. I. Soil water content gradients near a plane of onion roots. *J. Appl. Ecol.* 10, 585–598.
- Grzebisz, W., Floris, J., Van Noordwijk, M. (1989): Loss of dry matter and cell contents from fibrous roots of sugar beet due to sampling, storage and washing. *Plant Soil* 113, 53–57.
- Hacke, U. G., Sperry, J. S., Ewers, B. E., Ellsworth, D. S., Schäfer, K. V. R., Oren, R. (2000): Influence of soil porosity on water use in *Pinus taeda*. *Oecologia* 124, 495–505.
- Hao, X., De Jong, E. (1988): Growth of wheat and barley seedlings at different matric and osmotic potentials. *Agron. J.* 80, 807–811.

- Hare, P. D., Cress, W. A., Van Staden, J. (1998): Dissecting the role of osmolyte during stress. *Plant Cell Environ.* 21, 535–553.
- Hillel, D. (1998): *Environmental Soil Physics*. Academic Press, London, p. 771.
- Hinsinger, P., Gobran, G. R., Gregory, P. J., Wenzel, W. W. (2005): Rhizosphere geometry and heterogeneity arising from root-mediated physical and chemical processes. *New Phytologist* 168, 293–303.
- Homae, M. (1999): Root water uptake under non-uniform transient salinity and water stress. PhD thesis, Wageningen Agricultural University, The Netherlands.
- Hsiao, T. C., Läuchli, A. (1986): Role of potassium in plant-water relations, in Tinker, B.: *Advances in Plant Nutrition*. Springer, New York, pp. 281–312.
- Jackson, R. B., Sperry, J. S., Dawson, T. E. (2000): Root water uptake and transport: using physiological processes in global predictions. *Trends Plant Sci.* 5, 482–488.
- Kelly, S. F., Selker, J. S. (2001): Osmotically driven water vapour transport in unsaturated soils. *Soil Sci. Soc. Am. J.* 65, 1634–1641.
- Kirk, G. J. D. (1999): A model of phosphate solubilization by organic anion excretion from plant root. *Eur. J. Soil Sci.* 50, 369–378.
- Klute, A., Dirksen, C. (1986): Hydraulic Conductivity and Diffusivity: Laboratory methods, in Klute, A.: *Methods of Soil Analysis, Part 1: Physical and Mineralogical Methods*. American Society of Agronomy, Madison.
- Kramer, P. J., Boyer, J. S. (1995): *Water Relations of Plants and Soils*. Academic Press, Cambridge, p. 495.
- Li, Y., Xu, H., Cohen, S. (2005): Long-term hydraulic acclimation to soil texture and radiation load in cotton. *Plant, Cell Environ.* 28, 492–499.
- Martinez, J.-P., Lutts, S., Schanck, A., Bajji, M., Kinet, J.-M. (2004): Is osmotic adjustment required for water stress resistance in the Mediterranean shrub *Atriplex halimus* L? *J. Plant Physiol.* 161, 1041–1051.
- Mengel, K., Arneke, W.-W. (1982): Effect of potassium on the water potential, the pressure potential, the osmotic potential and cell elongation in leaves of *Phaseolus vulgaris*. *Physiol. Plant.* 54, 402–408.
- Morgan, J. M. (1984): Osmoregulation and water stress in higher plants. *Annu. Rev. Plant Physiol.* 35, 299–319.
- Mualem, Y. (1976): A new model for predicting the hydraulic conductivity of unsaturated porous media. *Water Resour. Res.* 12, 513–522.
- Nobel, P. S. (1999): *Physicochemical and Environmental Plant Physiology*. Academic Press, San Diego, p. 367.
- Peoples, T. R., Koch, D. W. (1979): Role of potassium in carbon dioxide assimilation in *Medicago sativa* L. *Plant Physiol.* 63, 878–881.
- Robin, C., Shamsun-Noor, L., Guckert, A. (1989): Effect of potassium on the tolerance to PEG-induced water stress of two white clover varieties (*Trifolium repens* L.). *Plant Soil* 120, 153–158.
- Schaap, M. G., Leij, F. J., Van Genuchten, M. Th. (2001): ROSETTA: a computer program for estimating soil hydraulic parameters with hierarchical pedotransfer functions. *J. Hydrol.* 251, 163–176.
- Seiffert, S., Kaselowsky, J., Jungk, A., Claassen, N. (1995): Observed and calculated potassium uptake by maize as affected by soil water content and bulk density. *Agron. J.* 87, 1070–1077.
- So, H. B., Nye, P. H. (1989): The effect of bulk density, water content and soil type on the diffusion of chloride in soil. *J. Soil Sci.* 40, 743–749.
- Sonneveld, C., Van den Bos, A. L., Voogt, W. (2004): Modeling osmotic salinity effects on yield characteristics of substrate-grown greenhouse crops. *J. Plant Nutr.* 27, 1931–1951.
- Stirzaker, R. J., Passioura, J. B. (1996): The water relations of the root-soil interface. *Plant, Cell Environ.* 19, 201–208.
- Van Genuchten, M. Th. (1980): A closed-form equation for predicting the hydraulic conductivity of unsaturated soils. *Soil Sci. Soc. Am. J.* 44, 892–898.
- Vetterlein, D., Jahn, R. (2004a): Combination of micro suction cups and time-domain reflectometry to measure osmotic potential gradients between bulk soil and rhizosphere at high resolution in time and space. *Eur. J. Soil Sci.* 55, 497–504.
- Vetterlein, D., Jahn, R. (2004b): Gradients in soil solution composition between bulk soil and rhizosphere – *In situ* measurement with changing water content. *Plant Soil* 258, 307–317.
- Vetterlein, D., Kuhn, K., Schubert, S., Jahn, R. (2004): Consequences of sodium exclusion for the osmotic potential in the rhizosphere – Comparison of two maize cultivars differing in Na⁺ uptake. *J. Plant Nutr. Soil Sci.* 167, 337–344.
- Wadleigh, C. H., Ayers, A. D. (1945): Growth and biochemical composition of bean plants as conditioned by soil moisture tension and salt concentration. *Plant Physiol.* 20, 106–132.
- Wenzel, W. W., Wieshammer, G., Fitz, W. J., Puschenreiter, M. (2001): Novel rhizobox design to assess rhizosphere characteristics at high spatial resolution. *Plant Soil* 237, 37–45.